

Chapter 3 Study Sheet

Parallel and Perpendicular Lines — everything you need on one page.

3.1 Pairs of Lines and Angles

- Parallel lines: coplanar, never intersect ($m \parallel n$). Perpendicular lines: meet at a right angle ($m \perp n$).
- Skew lines: do NOT intersect and are NOT coplanar. “Not intersecting” alone does not mean parallel.
- A transversal cuts two or more coplanar lines at different points, creating 8 angles.
- Corresponding (matching corners) · Alternate interior (between, opposite sides) · Alternate exterior (outside, opposite sides) · Consecutive interior (between, same side).

3.2 Parallel Lines and Transversals (lines ARE parallel → find angles)

- Corresponding Angles Postulate: corresponding angles are CONGRUENT.
- Alternate Interior Angles Theorem: alternate interior angles are CONGRUENT.
- Alternate Exterior Angles Theorem: alternate exterior angles are CONGRUENT.
- Consecutive Interior Angles Theorem: consecutive interior angles are SUPPLEMENTARY (sum to 180°).
- Shortcut: three of the four are congruent; only the same-side (consecutive) pair adds to 180° .

3.3 Proving Lines Parallel (angles are given → prove parallel)

- Every 3.2 theorem has a converse: congruent corresponding, alternate interior, or alternate exterior angles $\implies$ the lines are parallel.
- Supplementary consecutive interior angles $\implies$ the lines are parallel.
- Transitive Property of Parallel Lines: if $p \parallel q$ and $q \parallel r$, then $p \parallel r$.
- Watch out: congruent CONSECUTIVE interior angles prove nothing unless each is 90° .

3.4 Proofs with Perpendicular Lines

- The distance from a point to a line is the length of the PERPENDICULAR segment — it is the shortest one.
- A perpendicular bisector is perpendicular to a segment at its midpoint.
- Linear Pair Perpendicular Theorem: a linear pair of congruent angles $\implies$ the lines are perpendicular (each angle is 90°).
- Perpendicular Transversal Theorem: $t \perp$ one of two parallel lines $\implies t \perp$ the other.
- Lines Perpendicular to a Transversal Theorem: two lines $\perp$ the same line $\implies$ those two lines are parallel.

3.5 Equations of Parallel and Perpendicular Lines

- Slope: $m = (y_2 - y_1) \div (x_2 - x_1)$. Horizontal $\rightarrow m = 0$. Vertical $\rightarrow m$ is undefined.
- Parallel $\iff$ SAME slope, different y-intercepts. Perpendicular $\iff$ OPPOSITE RECIPROCAL slopes (product = -1).
- Coinciding $\iff$ same slope AND same y-intercept (one line written two ways).
- Point-slope: $y - y_1 = m(x - x_1)$. Slope-intercept: $y = mx + b$. Vertical: $x = a$. Horizontal: $y = b$.
- Always check: substitute the given point back into your final equation.